

O > 26
C 17-25
NW 10-16
ND < 10

COMPREHENSION
O > 10
C 6-9
NW 3-5
ND < 3

MODIFIED!
O > 15
C 9-14
NW 5-8
ND < 5

W-12

YEAR 9 SCIENCE
EARTH AND SPACE SCIENCES
TEST

NAME: SOLUTIONS FORM: _____ MARK: 35

Achievement Standard: students explain global features and events in terms of geological processes and timescales.

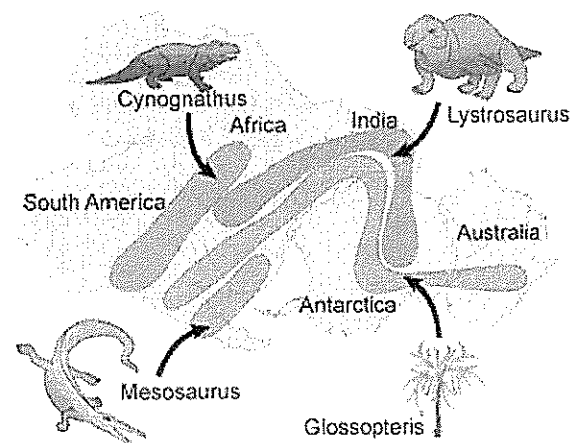
Part 1: Multiple Choice

Choose the correct answer for each question and write its letter in the appropriate box.

1. Which of the following statements is *correct*?
 - (a) Tsunamis involve a lot of moving water, but not a lot of energy.
 - (b) Tsunamis are only dangerous when they reach a coastline.
 - (c) Tsunamis are caused by underwater volcanoes.
 - (d) Tsunamis are caused by tides.
2. Which of the following statements is *correct*?
 - (a) The boundaries of the Earth's crustal plates are regions of great stability.
 - (b) Volcanoes can only erupt on the land, where they form large craters.
 - (c) Volcanoes are places on the Earth's crust where molten material from inside the Earth comes to the surface.
 - (d) Earthquakes are caused by volcanoes erupting.
3. Earthquakes and volcanoes are often found together because:
 - (a) they are both produced by movement of crustal plates.
 - (b) earthquakes cause volcanoes.
 - (c) both are formed in the same way.
 - (d) volcanoes cause earthquakes.
4. What is meant by the term 'magma'?
 - (a) Magma is a rock that has solidified from molten rock.
 - (b) Magma is molten rock that has flowed out onto the Earth's surface.
 - (c) Magma is molten rock under the surface of the Earth.
 - (d) Magma is an igneous rock such as granite.
5. The inside of the Earth consists of four major layers. Which is the hottest layer?
 - (a) mantle
 - (b) outer core
 - (c) crust
 - (d) inner core
6. Which layer of the Earth causes the movement of the continents?
 - (a) mantle
 - (b) outer core
 - (c) crust
 - (d) inner core

MULTIPLE CHOICE ANSWERS	
1	B
2	C
3	A
4	C
5	D
6	A
7	B
8	D
9	B
10	C
11	B
12	D

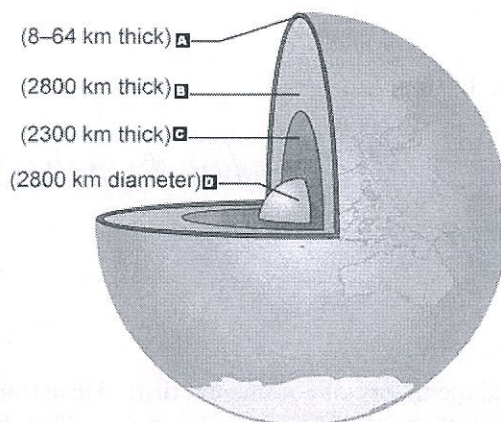
7. Which of the following statements is *correct*?
- Volcanoes can only erupt on the land, where they form large craters.
 - Volcanoes are places on the Earth's crust where molten material from inside the Earth comes to the surface.
 - Earthquakes are caused by volcanoes erupting.
 - The boundaries of the Earth's crustal plates are regions of great stability.
8. Areas of the Earth where new crust is being formed are called:
- continental plates.
 - subduction zones.
 - ocean trenches.
 - mid-ocean ridges.
9. What occurs at a subduction zone?
- An oceanic plate is rising to the surface and forming new land.
 - An oceanic plate is submerging beneath a continental plate.
 - A continental plate is submerging beneath an oceanic plate.
 - The crustal plates are moving apart and forming new land.
10. The Earth's crust is broken into many plates that appear to float and move. Explain this movement.
- Over millions of years, ice ages have caused different temperatures that allow convection of water currents.
 - The oceans cause weathering and erosion, which makes plates move.
 - The asthenosphere is molten and develops convection currents that move the plates.
 - Magnetic fields cause the plates to be attracted towards the poles.
11. Alfred Wegener proposed a theory in which the continents split apart and moved. Name this original supercontinent.
- Laurasia
 - Pangaea
 - Gondwana
 - Wegenerville
12. This diagram shows fossil evidence to support the idea that the continents were once joined. Identify the organism not found across South America.



- Mesosaurus
- Glossopteris
- Cynognathus
- Lystrosaurus

Part 2: Short Answer

1. Label the following diagram showing the layers of the Earth, giving the state (solid, liquid, etc) of the material found in each.



	Name of layer	State (solid, liquid, etc)
A	crust	solid
B	mantle	plastic
C	outer core	liquid
D	inner core	solid

(4 marks)

2. In the table below, choose the correct statement for each word given. **Write the letter of the correct statement next to the word.**

(4 marks)

WORD	CORRECT LETTER	STATEMENT
magma	F	A - how an organism looks due to its genes. movement in hot liquids caused by the rising of hot liquids.
plate tectonics	D	B - one tectonic plate moves under another plate.
sea-floor spreading	G	C - formed by two plates sliding past each other.
convection current	A	D - theory that large plates of the Earth gradually move.
transform boundary	C	E - theory that continents move through oceanic crust.
subduction	B	F - hot liquid rock that comes up from the mantle.
converging boundary	H	G - theory that the middle of the ocean is spreading apart.
continental drift	E	H - formed when two plates move towards each other.

3. (a) Where on the surface of the Earth are most volcanoes found?

Edges of tectonic plates.

(1 mark)

- (b) Explain why this is the case.

The plates are moving relative to each other, putting pressure on the magma.

(1 mark)

4. Alfred Wegener pioneered the theory of continental drift. He assumed that all of the continents fitted together in a "super continent" about 220 million years ago. They broke apart and started to drift slowly apart.

- (a) What name did he give to this "super continent"?

Pangaea

(1 mark)

- (b) What evidence did Wegener use to support his theory? Give two points.

• *coastline fit*

• *similar fossils in different continents*

• *rocks + landforms formed by glaciers are similar.*

(ANY 2)

(2 marks)

- (c) Explain what provides the force for the continental plates to move and drift.

Convection currents. (1)

Hot magma rises and moves along the crustal plates, causing them to move. (1)

(2 marks)

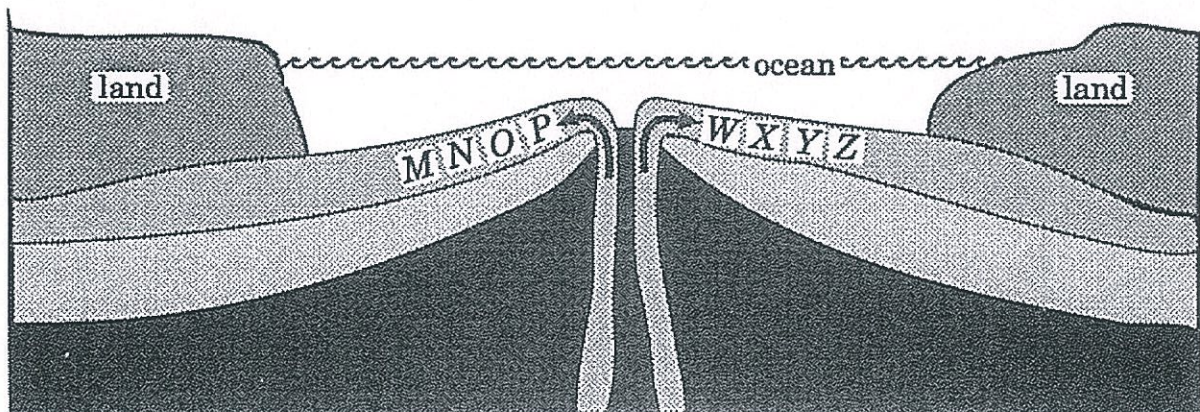
5. There are *three possible plate combinations at converging boundaries*. List the three combinations and explain what landforms they form.

- (i) OCEAN TO CONTINENT - forms a subduction zone, forming a trench.
(1) (mountains along edge of continent OK). (1)
- (ii) CONTINENT TO CONTINENT - push up mountains. (1)
(1)
- (iii) OCEAN TO OCEAN - one pushes below the other, forming a trench.
(1) (1)

(6 marks)

6. In a **spreading ridge** the floor of the ocean moves out equally in opposite directions. Look at the following 'side-on' view of a spreading ridge. Which set of rocks would have the same age?

MZ NY OX PW



(2 marks)

AFTER YOU HAVE FINISHED THIS SECTION OF THE TEST MAKE SURE YOU CHECK YOUR ANSWERS CAREFULLY AGAIN AND THEN HAND IT INTO THE TEACHER. YOU WILL THEN BE GIVEN THE THIRD SECTION OF THE TEST TO COMPLETE.

Part C – Comprehension of a scientific piece of writing.

Plate tectonics may grind to a halt, then start again

ScienceDaily (7 January 2008)

Plate tectonics, the geologic process responsible for creating the Earth's continents, mountain ranges, and ocean basins, may be an on-again, off-again affair.

Scientists have assumed that the shifting of crustal plates has been slow but continuous over most of the Earth's history, but a new study from researchers at the Carnegie Institution suggests that plate tectonics may have ground to a halt at least once in our planet's history—and may do so again.

A key aspect of plate tectonic theory is that on geologic time scales ocean basins are transient features, opening and closing as plates shift. Basins are consumed by a process called subduction, where oceanic plates descend into the Earth's mantle. Subduction zones are the sites of oceanic trenches, high earthquake activity and most of the world's major volcanoes.

Writing in the 4 January issue of *Science*, Paul Silver of the Carnegie Institution's Department of Terrestrial Magnetism and former postdoctoral fellow Mark Behn (now at Woods Hole Oceanographic Institution) points out that most of today's subduction zones are located in the Pacific Ocean basin. If the Pacific basin were to close, as it is predicted to do about in 350 million years when the westward-moving Americas collide with Eurasia, then most of the planet's subduction zones would disappear with it.

This would effectively stop plate tectonics unless new subduction zones start up, but subduction initiation is poorly understood. 'The collision of India and Africa with Eurasia between 30 and 50 million

years ago closed an ocean basin known as Tethys,' says Silver. 'But no new subduction zones have initiated south of either India or Africa to compensate for the loss of subduction by this ocean closure.'

Silver and Behn also present geochemical evidence from ancient igneous rocks indicating that around one billion years ago there was a lull in the type of volcanic activity normally associated with subduction. This idea fits with other geologic evidence for the closure of a Pacific-type ocean basin at that time, welding the continents into a single 'supercontinent' (known to geologists as Rodinia) and possibly snuffing out subduction for a while. Rodinia eventually split apart when subduction and plate tectonics resumed.

Plate tectonics is driven by heat flowing from the Earth's interior, and a stoppage would slow the rate of the Earth's cooling, just as clamping a lid on a soup pot would slow the soup's cooling. By periodically clamping the lid on heat flow, intermittent plate tectonics may explain why the Earth has lost heat slower than current models predict. And the build-up of heat beneath stagnant plates may explain the occurrence of certain igneous rocks in the middle of continents away from their normal locations in subduction zones.

'If plate tectonics indeed starts and stops, then continental evolution must be viewed in an entirely new light, since it dramatically broadens the range of possible evolutionary scenarios,' says Silver.

Answer the following questions in the Blog set up for your class on the Science Year 9 Moodle page.

1. What assumption have scientists made about plate tectonics that may not be true?
2. What geological process consumes ocean basins?
3. What are the common features of subduction zones?
4. Where are most of the world's subduction zones?
5. Does the loss of the Earth's heat provide evidence for continual motion of the plates or stop-start motion?
6. Find out about the supercontinent Rodinia – did it come before or after Pangaea?

READING COMPREHENSION

1. Movement is slow and continuous.
Evidence that it may have stopped. (2)
2. Subduction - oceanic plates descend into the Earth. (1)
3. Sites of oceanic trenches, high earthquake activity, most of the major volcanoes. (2)
4. Located in the Pacific Ocean basin. (1)
5. Stop-start. Trapping heat slows the cooling of the Earth. (2)
6. Rodinia - formed 1.1 → 1.2 billion years ago (depending on information source).
- maybe as early as 750 million years ago.

Pangaea - formed 250-300 million years ago (depending on information source).

Therefore - Rodinia formed first.

Rodinia - composed of parts of North America, South America, Australia and Antarctica.

- broke up due to the internal heating of the Earth.
- ultimately the broken up parts of Rodinia came together to form Pangaea.

(5)

